Model Averaging Approach for Analyzing Complex Survey Data

Abstract

Survey data are widely used across various fields. Model-assisted inference provides a useful framework for analyzing such data by incorporating auxiliary information through a superpopulation model to improve estimation precision. However, accurately specifying the superpopulation model in practice is challenging. To account for uncertainty in model specification, this paper develops a novel model averaging framework for analyzing complex survey data, in which the weights are determined by minimizing a Mallows-type criterion. The resulting model averaging estimator is shown to be asymptotically optimal in the sense of achieving the minimum possible loss. We also establish the convergence rate of the estimated weight vector toward the theoretically optimal weight vector. When the set of candidate models includes one or more correct models, the sum of the estimated weights assigned to these models is shown to converge to one. Based on the proposed model averaging approach, we further construct a model-assisted averaging estimator of the finite population total and establish its asymptotic normality. We also derive its mean squared error and provide a consistent estimator of this quantity. Numerical analyses demonstrate the desirable finite-sample performance of our new methods.

Keywords: Finite population, Mallows-type criterion, Model-assisted inference, Superpopulation model

1 Introduction

Survey data provide valuable information on population characteristics and behaviors, thereby supporting decision-making and policy development. For example, companies use survey data to understand consumer needs and preferences, which helps inform product development and marketing strategies. Similarly, governments conduct public opinion surveys to formulate new policies and improve public services. In survey sampling, three inferential frameworks are commonly used to analyze complex survey data: design-based, model-based, and model-assisted inference. Design-based inference, also known as randomization inference, is the foundation of survey sampling theory (Cochran 1977). Under this framework, the finite population is treated as fixed, and all randomness with the estimators is attributed solely to the sampling design. Both model-based and model-assisted inference use a superpopulation model to characterize the relationship between the study variable and the auxiliary variables. Under model-based inference, the finite population is regarded as a realization from a superpopulation model, and its estimation can be formulated as a prediction problem for nonsampled units without consideration of randomness of sampling. The resulting estimators are assessed totally with respect to the working model (Valliant et al. 2000). Model-assisted inference combines information from the sampling design and the assumed model to construct finite population estimators, while typically focusing on their design-based properties (Särndal et al. 1992). A number of studies have compared these three inferential approaches (Särndal et al. 1978, Särndal 1980, Hansen et al. 1983, Gregoire 1998). In general, design-based estimators do not rely on any model and are therefore robust to model misspecification, whereas model-based estimators are efficient when the assumed model is correctly specified but may fail under severe misspecification. By contrast, model-assisted estimators combine the strengths of both approaches. Under certain regularity conditions, they retain design consistency even when the model is misspecified.

When the model is correctly specified, they can be more efficient than purely design-based estimators such as the Horvitz–Thompson estimator (Horvitz & Thompson 1952).

To date, many model-assisted estimators have been developed based on various superpopulation models. The model-assisted estimation based on linear models, commonly known as the generalized regression (GREG) estimation (Cassel et al. 1976), has been extensively studied (Cassel et al. 1977, Särndal 1980, Särndal et al. 1992). Moving beyond parametric specifications, Breidt & Opsomer (2000) employed local polynomial regression to construct a model-assisted estimator under a nonparametric model, while Breidt et al. (2007) subsequently proposed a semiparametric counterpart. Wang (2009) and Wang & Wang (2011) further extended the model-assisted framework to single-index and nonparametric additive models, respectively. For a high-dimensional linear model, Ta et al. (2020) proposed a GREG estimator based on ℓ_1-penalized weighted least squares. Despite these developments, existing studies generally rely on a prespecified superpopulation model and do not account for model uncertainty. If the assumed model is misspecified, the efficiency gains from model-assisted inference may be diminished. One way to address this issue is model averaging, which forms a weighted average of estimators/forecasts obtained from a number of candidate models for inference on the target of interest (Claeskens & Hjort 2008). Unlike model selection methods, which base inference on a single "winning" model, model averaging leverages information from all candidate models and avoids selecting a poorly performing model, thereby potentially reducing estimation risk and improving estimation accuracy and stability (Leung & Barron 2006, Wan et al. 2010).

Model averaging can be classified into Bayesian model averaging (BMA) and frequentist model averaging (FMA). Since the prior specifications required by BMA may be mutually conflicting and difficult to interpret in practice (Hjort & Claeskens 2003), this paper focuses on FMA. A central issue in FMA is weight selection for the candidate models, which has

received considerable attention in the literature. Based on classical information criteria, smoothed AIC and smoothed BIC have been suggested (c.f., Buckland et al. 1997). Yang (2001) proposed an adaptive model averaging method. Building on Mallows' criterion, Hansen (2007) developed Mallows model averaging (MMA) for the nested candidate models and established the asymptotic optimality of the resulting estimator. Wan et al. (2010) further provided an improved proof of asymptotic optimality for MMA under a non-nested model framework. Liang et al. (2011) proposed optimal model averaging by minimizing an unbiased estimator of the mean squared error, while Hansen & Racine (2012) introduced jackknife model averaging (JMA) based on cross-validation. Zhang et al. (2015) proposed a model averaging method based on the Kullback-Leibler distance. These studies have played an important role in the development of FMA, which has subsequently been extended to a wide range of settings, including mixed-effect models (Zhang et al. 2014), generalized linear models (Zhang et al. 2016), semiparametric and nonparametric models (Li et al. 2018, Zhang & Wang 2019, Zhu et al. 2019), quantile regression (Lu & Su 2015), measurement error models (Wang et al. 2012, Zhang et al. 2019), high-dimensional data (Ando & Li 2014), missing data (Fang et al. 2019), functional data (Zhang et al. 2018), and time series data (Gao et al. 2016, Liao et al. 2019). Moreover, model averaging with nonnegative weights summing to one can be viewed as a form of convex aggregation; see Nemirovski (2000), Yang (2000), Tsybakov (2003) and Bunea et al. (2007) for studies on aggregation procedures.

In the context of survey sampling, Liang & Kuk (2004) and Clyde & Ghosh (2012) developed BMA approaches for finite population estimation, while Elliott (2008, 2009) used BMA to construct weight-trimming estimators. BMA has also been investigated in small area estimation (Aitkin et al. 2009). Within the FMA framework, Li & Opsomer (2006) combined model-assisted estimators constructed using different bandwidths to estimate the finite population total and examined the numerical performance of several weight selection

methods. For volunteer web surveys, Liu et al. (2023) estimated the finite population mean by combining two estimators constructed from two different propensity score models. However, neither the theoretical properties of the weight selection methods nor those of the resulting model averaging estimators have been fully established.

The purpose of this paper is to develop a novel model averaging framework for analyzing complex survey data. Specifically, we first construct a model averaging estimator of the mean function in the superpopulation model. A Mallows-type weight selection criterion is developed, and its asymptotic unbiasedness and consistency are established. We further derive the asymptotic optimality of the proposed method and the convergence rate of the weight estimator toward the theoretically optimal weight that minimizes the risk function. Moreover, when the set of candidate models contains one or more correct models (i.e., models that include the true model), the estimated weights are concentrated on these models asymptotically. These theoretical derivations not only allow the number of candidate models and the maximum model dimension to diverge, but also account for randomness in both the model and the sampling design, making the proofs technically challenging. On the other hand, finite population parameters, such as population total, mean, proportion, and distribution function (all of which can be expressed as functions of population totals), are of primary interest in survey sampling. We therefore construct a model-assisted averaging estimator of the population total based on the proposed MMA method. The resulting estimator is shown to be asymptotically normal, and its mean squared error and an asymptotically unbiased and consistent estimator thereof are derived. Simulation studies and real data applications demonstrate the effectiveness of the proposed methods.

The remainder of the paper is organized as follows. Section 2 introduces the notation and inference framework. Section 3 develops a model averaging method for superpopulation model based on survey data and investigates its asymptotic properties. Section 4 establishes

theoretical results for the proposed model-assisted averaging estimator of the finite population total. Section 5 presents numerical studies to demonstrate the effectiveness of the proposed methods. Section 6 concludes. Proofs of all theoretical results and additional numerical analyses are provided in the Supplementary Material.

2 Notation and inference framework

Consider a finite population $U = \{1, \ldots, N\}$. For each unit $i \in U$, the study variable takes the value y_i. A sample s of size n is drawn from U according to a sampling design $P(s)$, where $P(s)$ is the probability of drawing the sample s. For all $i, j \in U$, the first- and second-order inclusion probabilities are defined as $\pi_i = \sum_{\{s:\ s \ni i\}} P(s)$ and $\pi_{ij} = \sum_{\{s:\ s \ni i,j\}} P(s)$, respectively. In particular, $\pi_{ii} = \pi_i$. Let $I_i = 1$ if $i \in s$ and $I_i = 0$ otherwise. Then $E_d(I_i) = \pi_i$, where $E_d(\cdot)$ denotes expectation with respect to the sampling design, that is, averaging over all possible samples drawn from the finite population.

To estimate the population total $t_y = \sum_U y_i$, Horvitz & Thompson (1952) proposed the well-known Horvitz–Thompson (HT) estimator:

$$\hat{t}_{y\pi} = \sum_{i \in s} \frac{y_i}{\pi_i}.$$

This estimator is design-unbiased (i.e., $E_d(\hat{t}_{y\pi}) = t_y$), and its design-based variance is

$$V_d(\hat{t}_{y\pi}) = \sum_{i,j \in U} \frac{\pi_{ij} - \pi_i \pi_j}{\pi_i \pi_j} y_i y_j.$$

The HT estimator is purely design-based and does not use any auxiliary information. A natural idea is to incorporate available auxiliary information to improve estimation efficiency.

Suppose that the study variable y_i is related to finite-dimensional auxiliary vector $\boldsymbol{x}_i = (x_{i1}, \ldots, x_{ip})'$ and follows the linear superpopulation model:

$$y_i = \sum_{j=1}^{p} x_{ij} \beta_j + e_i, \qquad i \in U, \tag{1}$$

where $\{\beta_j\}$ are the unknown regression coefficients, and $\{e_i\}$ are independent random errors satisfying $E_\epsilon(e_i|\boldsymbol{x}_i) = 0$ and $V_\epsilon(e_i|\boldsymbol{x}_i) = \sigma^2$. This superpopulation model introduces an additional source of randomness, with $E_\epsilon(\cdot)$ and $V_\epsilon(\cdot)$ denoting the model-based expectation and variance, respectively.

To construct the model-assisted estimator of the population total, we need to estimate the unknown coefficients in the superpopulation model (1). Let $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)'$. Since the finite population $\{(y_i, \boldsymbol{x}_i), i \in U\}$ is a realization from the superpopulation model, the least squares estimator of $\boldsymbol{\beta}$ is $\tilde{\boldsymbol{\beta}} = (\boldsymbol{X}_U'\boldsymbol{X}_U)^{-1}\boldsymbol{X}_U'\boldsymbol{y}_U$, where $\boldsymbol{X}_U = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N)' \triangleq [\boldsymbol{x}_i]_{i\in U}$ and $\boldsymbol{y}_U = (y_1, \ldots, y_N)' \triangleq [y_i]_{i\in U}$. Then, a difference estimator of t_y can be written as

$$\hat{t}_{y,\text{diff}} = \sum_{i\in s} \frac{y_i - \tilde{\mu}_i}{\pi_i} + \sum_{i\in U} \tilde{\mu}_i,$$

where $\tilde{\mu}_i = \boldsymbol{x}_i'\tilde{\boldsymbol{\beta}}$. This difference estimator is design-unbiased, with the design-based variance

$$V_d(\hat{t}_{y,\text{diff}}) = \sum_{i,j\in U} \frac{\pi_{ij} - \pi_i\pi_j}{\pi_i\pi_j}(y_i - \tilde{\mu}_i)(y_j - \tilde{\mu}_j).$$

For further details on the difference estimator, see Section 6.3 of Särndal et al. (1992).

Note that $\tilde{\boldsymbol{\beta}}$ and $\tilde{\mu}_i$ are pseudo-estimators and cannot be computed in survey sampling, because the study variable is observed only for the sampled units. Based on the survey sample s, a design-weighted estimator of $\tilde{\boldsymbol{\beta}}$ is $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}_s'\boldsymbol{\Phi}_s\boldsymbol{X}_s)^{-1}\boldsymbol{X}_s'\boldsymbol{\Phi}_s\boldsymbol{y}_s$, where $\boldsymbol{y}_s = [y_i]_{i\in s}$, $\boldsymbol{X}_s = [\boldsymbol{x}_i]_{i\in s}$, and $\boldsymbol{\Phi}_s = \text{diag}\{\pi_i^{-1}\}_{i\in s}$. Accordingly, the model-assisted estimator of the population total t_y is given by

$$\hat{t}_y = \sum_{i\in s} \frac{y_i - \hat{\mu}_i}{\pi_i} + \sum_{i\in U} \hat{\mu}_i, \tag{2}$$

where $\hat{\mu}_i = \boldsymbol{x}_i'\hat{\boldsymbol{\beta}}$. Sections 5.10 and 6.6 of Särndal et al. (1992) discuss the design-based properties of $\hat{\boldsymbol{\beta}}$ and $\hat{t}_y$, respectively. However, the aforementioned studies generally rely on a prespecified superpopulation model. When the assumed model is misspecified, the resulting estimators may be unstable, and the efficiency gains from model-assisted inference

may be reduced. To address these limitations, we use the model averaging approach in the next section to estimate the superpopulation model.

3 Model averaging estimation for superpopulation model

In this section, we construct a model averaging estimator of the mean function in the superpopulation model. A Mallows-type criterion is developed to select the averaging weights, and the theoretical properties of the resulting estimator are established. To this end, we first introduce a more general superpopulation model. Suppose that the study variable y_i is related to the countably infinite-dimensional auxiliary vector $\boldsymbol{x}_i^\dagger = (x_{i1}, x_{i2}, \dots)'$ and follows the superpopulation model:

$$y_i = \sum_{j=1}^{\infty} x_{ij}\beta_j + e_i \triangleq \mu_i + e_i, \qquad i \in U, \tag{3}$$

where $\mu_i = \sum_{j=1}^{\infty} x_{ij}\beta_j$ is the mean function, $\{\beta_j\}$ are the unknown regression coefficients, and $\{e_i\}$ are independent random errors satisfying $E_\epsilon(e_i|\boldsymbol{x}_i^\dagger) = 0$ and $V_\epsilon(e_i|\boldsymbol{x}_i^\dagger) = \sigma^2$. The subsequent analysis is conducted under the superpopulation model (3).

Remark 3.1. In practice, the variable of interest may be influenced by infinitely many factors. Accordingly, we consider an infinite-dimensional auxiliary vector in the superpopulation model (3). This formulation includes the conventional finite-dimensional model (1) as a special case, in which all but finitely many regression coefficients are zero. Moreover, as noted by Hansen (2007), it accommodates nonparametric regression by allowing the covariates to represent basis functions in a series expansion. Similar formulations are common in the model averaging literature (Hansen 2007, Wan et al. 2010, Hansen & Racine 2012, Liao & Zou 2020). On the other hand, because human understanding remains limited, the (candidate) models used in practice are typically finite-dimensional. Hence, the model-assisted estimation theory for finite-dimensional models can be utilized.

3.1 Model averaging estimator of the mean function

To construct the model-assisted estimator of the population total, we first estimate the mean function in the superpopulation model based on the model averaging approach. Consider M candidate models used to approximate the superpopulation model (3), where M is allowed to increase with the sample size n. The mth approximating model employs k_m regressors selected from $\boldsymbol{x}_i$ and is given by

$$y_i = \sum_{j=1}^{k_m} x_{ij(m)}\beta_{j(m)} + e_i, \tag{4}$$

where $x_{ij(m)}$ denotes the jth regressor included in the mth model and $\beta_{j(m)}$ is the corresponding coefficient. Let $\boldsymbol{x}_{i(m)} = (x_{i1(m)}, \ldots, x_{ik_m(m)})'$. Define $\boldsymbol{X}_{(m)U} = [\boldsymbol{x}_{i(m)}]_{i\in U}$, $\boldsymbol{\beta}_{(m)} = (\beta_{1(m)}, \ldots, \beta_{k_m(m)})'$, and $\boldsymbol{e}_U = [e_i]_{i\in U}$. Then, the mth approximating model can be written as $\boldsymbol{y}_U = \boldsymbol{X}_{(m)U}\boldsymbol{\beta}_{(m)} + \boldsymbol{e}_U$. Based on the entire finite population, the least squares estimator of $\boldsymbol{\beta}_{(m)}$ is $\tilde{\boldsymbol{\beta}}_{(m)} = (\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U})^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{y}_U$.

Let $\boldsymbol{\omega} = (\omega_1, \ldots, \omega_M)'$ be a weight vector belonging to the simplex in $\mathbb{R}^M$, $\mathcal{W} = \{\boldsymbol{\omega} \in [0,1]^M : \sum_{m=1}^M \omega_m = 1\}$. Following Hansen (2007) and Wan et al. (2010), the model averaging estimator of the mean function μ_i is given by

$$\tilde{\mu}_i(\boldsymbol{\omega}) = \sum_{m=1}^M \omega_m \left(\boldsymbol{x}'_{i(m)}\tilde{\boldsymbol{\beta}}_{(m)}\right) \triangleq \sum_{m=1}^M \omega_m \tilde{\mu}_{i(m)}. \tag{5}$$

Note that $\tilde{\boldsymbol{\beta}}_{(m)}$ and $\tilde{\mu}_i(\boldsymbol{\omega})$ are pseudo-estimators and cannot be computed in survey sampling. Define $\boldsymbol{X}_{(m)s} = [\boldsymbol{x}_{i(m)}]_{i\in s}$. Then, based on the survey sample s, the design-weighted estimator of $\tilde{\boldsymbol{\beta}}_{(m)}$ is $\hat{\boldsymbol{\beta}}_{(m)} = (\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s})^{-1}\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{y}_s$ provided that $\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s}$ is invertible. Using $\hat{\boldsymbol{\beta}}_{(m)}$ in place of $\tilde{\boldsymbol{\beta}}_{(m)}$ in Equation (5), we obtain a survey-based model averaging (SMA) estimator of the mean function μ_i:

$$\hat{\mu}_i(\boldsymbol{\omega}) = \sum_{m=1}^M \omega_m \left(\boldsymbol{x}'_{i(m)}\hat{\boldsymbol{\beta}}_{(m)}\right) \triangleq \sum_{m=1}^M \omega_m \hat{\mu}_{i(m)}. \tag{6}$$

For a given weight vector $\boldsymbol{\omega}$, replacing $\hat{\mu}_i$ in the model-assisted estimator (2) with $\hat{\mu}_i(\boldsymbol{\omega})$ yields the following model-assisted averaging estimator of the population total t_y:

$$\hat{t}_y(\boldsymbol{\omega}) = \sum_{i\in s} \frac{y_i - \hat{\mu}_i(\boldsymbol{\omega})}{\pi_i} + \sum_{i\in U} \hat{\mu}_i(\boldsymbol{\omega}) \triangleq \sum_{m=1}^{M} \omega_m \hat{t}_{y(m)}, \tag{7}$$

where $\hat{t}_{y(m)} = \sum_s (y_i - \hat{\mu}_{i(m)})/\pi_i + \sum_U \hat{\mu}_{i(m)}$ is the model-assisted estimator under the mth candidate model. However, $\boldsymbol{\omega}$ is unknown in practice and needs to be estimated. We therefore develop a weight selection procedure in the next subsection. The theoretical properties of the model-assisted averaging estimator are established in Section 4.

Before proceeding, we introduce the notational conventions used throughout the paper. Population-level vectors and matrices are denoted by symbols like $\boldsymbol{y}_U$, $\boldsymbol{e}_U$, and $\boldsymbol{X}_{(m)U}$, with the corresponding sample quantities denoted by symbols like $\boldsymbol{y}_s$, $\boldsymbol{e}_s$, and $\boldsymbol{X}_{(m)s}$, respectively. A tilde is used for pseudo-estimators, such as $\tilde{\boldsymbol{\beta}}_{(m)}$, $\tilde{\mu}_{i(m)}$, and $\tilde{\mu}_i(\boldsymbol{\omega})$, whereas a hat denotes their sample-based counterparts, such as $\hat{\boldsymbol{\beta}}_{(m)}$, $\hat{\mu}_{i(m)}$, and $\hat{\mu}_i(\boldsymbol{\omega})$.

3.2 Weight selection criterion

Define the loss function as $L_N(\boldsymbol{\omega}) = (\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}))'(\boldsymbol{\mu}_U - \hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}))$ and the corresponding risk function as $R_N(\boldsymbol{\omega}) = E(L_N(\boldsymbol{\omega})|\boldsymbol{x}_1^\dagger, \ldots, \boldsymbol{x}_N^\dagger)$, where $\boldsymbol{\mu}_U = [\mu_i]_{i\in U}$, $\hat{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) = [\hat{\mu}_i(\boldsymbol{\omega})]_{i\in U}$, and $E(\cdot) = E_\epsilon(E_d(\cdot))$. The theoretically optimal weight vector is then defined as $\boldsymbol{\omega}^* = \arg\min_{\boldsymbol{\omega}\in\mathcal{W}} \{R_N(\boldsymbol{\omega})\}$, and the minimum risk is $\xi_N = \inf_{\boldsymbol{\omega}\in\mathcal{W}} R_N(\boldsymbol{\omega})$. A natural approach to estimating $\boldsymbol{\omega}^*$ is to minimize an appropriate estimator of the risk function. The Mallows criterion proposed by Hansen (2007) provides an unbiased estimator of the target risk up to an additive constant. In the same spirit, we propose the following Mallows-type criterion based on the survey sample s for selecting the weight vector:

$$C_n(\boldsymbol{\omega}) = \Big(\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\Big)' \boldsymbol{\Phi}_s \Big(\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\Big) + 2f^{-1}\sigma^2 k(\boldsymbol{\omega}) \tag{8}$$

where $\hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega}) = [\hat{\mu}_i(\boldsymbol{\omega})]_{i\in s}$, $f = n/N$, and $k(\boldsymbol{\omega}) = \sum_{m=1}^{M} \omega_m k_m$. The estimated weight vector is then defined as

$$\hat{\boldsymbol{\omega}} = (\hat{\omega}_1, \ldots, \hat{\omega}_M)' = \arg\min_{\boldsymbol{\omega}\in\mathcal{W}} \{C_n(\boldsymbol{\omega})\}. \tag{9}$$

Remark 3.2. Like other Mallows-type criteria, $C_n(\boldsymbol{\omega})$ consists of a goodness-of-fit term and a penalty term. However, both components are adjusted for the sampling design through the inclusion probability matrix $\boldsymbol{\Phi}_s$ and the sampling fraction f. Intuitively, the penalty term $2f^{-1}\sigma^2 k(\boldsymbol{\omega})$ in criterion (8) restricts the number and dimensions of the candidate models and guards against overfitting.

Remark 3.3. The motivation for our criterion is to approximate the risk function up to an additive constant. Based on this principle, we derive the penalty term $2\sigma^2 k_s(\boldsymbol{\omega})$, where $k_s(\boldsymbol{\omega}) = \sum_{m=1}^{M} \omega_m \mathrm{tr}\{\boldsymbol{\Phi}_s \boldsymbol{X}_{(m)s}(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\boldsymbol{X}_{(m)s})^{-1}\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}_s\}$. The derivation relies on the interchangeability condition $E_\epsilon E_d(\cdot) = E_d E_\epsilon(\cdot)$, which generally holds but may fail under an informative sampling design, a design associated with the study variable (Särndal et al. 1978, Raghunath 1990). Simplifying this penalty term yields the weight selection criterion (8). Detailed derivations and discussions are provided in Supplementary Material S1. Under the same conditions, all the theoretical results established in this paper remain valid regardless of which of the two penalty terms is used. Our numerical studies also compare the two criteria and show that they yield similar finite-sample performance.

Remark 3.3. The proposed criterion (8) reduces to the Mallows criterion of Hansen (2007) when the entire finite population is observed (i.e., $n = N$ and $\boldsymbol{\Phi}_s = \boldsymbol{I}_n$). Moreover, under simple random sampling (SRS), the proposed criterion simplifies to

$$C_n^{\mathrm{SRS}}(\boldsymbol{\omega}) = \big(\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\big)'\big(\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\big) + 2\sigma^2 k(\boldsymbol{\omega}),$$

which coincides with the criterion based on an independent sample.

In practice, the variance σ^2 is typically unknown. We estimate it using the largest

candidate model (assumed to be the Mth model), and denote the resulting estimator by

$$\hat{\sigma}^2_{(M)} = \frac{1}{n - k_M}\Big(\boldsymbol{y}_s - \boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\Big)'\Big(\boldsymbol{y}_s - \boldsymbol{X}_{(M)s}\hat{\boldsymbol{\beta}}_{(M)}\Big).$$

This estimator is model-unbiased for σ^2 when the Mth model is correctly specified. Replacing σ^2 in criterion (8) with $\hat{\sigma}^2_{(M)}$ yields the feasible Mallows-type criterion:

$$C^*_n(\boldsymbol{\omega}) = \Big(\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\Big)'\boldsymbol{\Phi}_s\Big(\boldsymbol{y}_s - \hat{\boldsymbol{\mu}}_s(\boldsymbol{\omega})\Big) + 2f^{-1}\hat{\sigma}^2_{(M)}k(\boldsymbol{\omega}). \tag{10}$$

Accordingly, a feasible estimator of the optimal weight vector is defined as

$$\hat{\boldsymbol{\omega}}^* = (\hat{\omega}^*_1, \ldots, \hat{\omega}_M)' = \underset{\boldsymbol{\omega}\in\mathcal{W}}{\arg\min}\left\{C^*_n(\boldsymbol{\omega})\right\}. \tag{11}$$

Note that both (9) and (11) correspond to standard quadratic programming problems. Therefore, $\hat{\boldsymbol{\omega}}$ and $\hat{\boldsymbol{\omega}}^*$ can be efficiently computed using widely available statistical or numerical software.

3.3 Theoretical properties

In this subsection, we establish the theoretical properties of the proposed model averaging method for the mean function in the superpopulaton model. Since the resulting model averaging estimator is subject to randomness from both the superpopulation model and the sampling design, we conduct our analysis under the joint model–design distribution, following Cassel et al. (1976), Särndal et al. (1978), Särndal (1980), and Isaki & Fuller (1982). Throughout the paper, all asymptotic results are established as N, $n \to \infty$. We first introduce additional notation and state the regularity conditions required for the subsequent theoretical analysis.

Based on the pseudo-estimator $\tilde{\mu}_i(\boldsymbol{\omega})$, we define the loss function $\tilde{L}_N(\boldsymbol{\omega}) = (\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}))'(\boldsymbol{\mu}_U - \tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}))$ and the corresponding risk function $\tilde{R}_N(\boldsymbol{\omega}) = E_\epsilon(\tilde{L}_N(\boldsymbol{\omega})|\boldsymbol{x}^\dagger_1, \ldots, \boldsymbol{x}^\dagger_N)$, where $\tilde{\boldsymbol{\mu}}_U(\boldsymbol{\omega}) = [\tilde{\mu}_i(\boldsymbol{\omega})]_{i\in U}$. Let $\tilde{\xi}_N = \inf_{\boldsymbol{\omega}\in\mathcal{W}}\{\tilde{R}_N(\boldsymbol{\omega})\}$. For the mth candidate model, define $\mu^*_{i(m)} = \boldsymbol{x}'_{i(m)}\boldsymbol{\beta}^*_{(m)}$ and $e^*_{i(m)} = y_i - \mu^*_{i(m)}$, where $\boldsymbol{\beta}^*_{(m)} = (\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U})^{-1}\boldsymbol{X}'_{(m)U}\boldsymbol{\mu}_U$.

Condition 1 *$E_\epsilon(e^{*4}_{i(m)})$ and $|x_{ij(m)}|$ are uniformly bounded in i, j and m. There exist positive constants c_1, c_2, c_3, and c_4 such that*

$$c_1 \leq \lambda_{\min}(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}/N) \leq \lambda_{\max}(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}/N) \leq c_2$$

and

$$c_3 \leq \lambda_{\min}(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}\boldsymbol{X}_{(m)s}/N) \leq \lambda_{\max}(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}\boldsymbol{X}_{(m)s}/N) \leq c_4,$$

where $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote the smallest and largest eigenvalues of a matrix, respectively.

Condition 2 $\min_{i\in U} \pi_i \geq c_5 > 0$, $\min_{i,j\in U} \pi_{ij} \geq c_6 > 0$, *and*

$$\limsup_{n\to\infty} n \max_{i\neq j\in U} \mid \pi_{ij} - \pi_i\pi_j \mid < \infty,$$

where c_5 and c_6 are positive constants.

Condition 3 $M^2\sqrt{nk_M}(f\xi_N)^{-1} = o(1)$.

Condition 1 imposes assumptions on the residuals, covariates, and design matrices of all candidate models. This condition is generally mild but may not hold when the errors or covariates in the candidate models are strongly heterogeneous. Similar conditions can be found in Zong et al. (2019, 2026) and Liao et al. (2019). Note that if both $\{k_m\}$ and $|x_{ij(m)}|$ are bounded, then $\lambda_{\max}(\boldsymbol{X}'_{(m)U}\boldsymbol{X}_{(m)U}/N) \leq c_2$ follows immediately. Further, $\lambda_{\max}(\boldsymbol{X}'_{(m)s}\boldsymbol{\Phi}\boldsymbol{X}_{(m)s}/N) \leq c_4$ also holds when the first-order inclusion probabilities are bounded away from zero. Condition 2 is a standard assumption on the first- and second-order inclusion probabilities in survey sampling. The final part of this condition captures the pairwise dependence among the sampled units (Robinson & Särndal 1983), which is satisfied under simple random sampling without replacement, Poisson sampling, and rejective sampling (Hájek 1964, Boistard et al. 2012), but may fail under designs with strong dependence, such as multistage or systematic sampling (Delevoye & Sävje 2020). Condition 3 characterizes the relationship among the sample size n, the number of candidate

models M, the maximum model dimension k_M, and the minimum risk ξ_N. Analogous to the minimum-risk assumption in Hansen (2007), this condition requires $f\xi_N \to \infty$, reflecting a certain degree of model misspecification. It also constrains the growth rates of M and k_M, requiring that neither increase too rapidly. Similar conditions can be found in Wan et al. (2010), Zhang et al. (2013), Zhang et al. (2016), and Liao et al. (2019). To further clarify Condition 3, we consider a simple setting in which the entire finite population is observed ($n = N$) and the mth approximating model contains the first m regressors ($M = k_M$). Let $M = O(N^v)$ for some $v > 0$. Suppose that the risk of the mth candidate model is of order $Nm^{-a} + m$, where $a > 0$. This order coincides with the risk rate of the series estimator given in Theorem 1 of Newey (1997). See also Wan et al. (2010) and Liao et al. (2019). It follows that $\xi_N = O(N^{1-av} + N^v)$. Combining this result with Equation (A.14) of Wan et al. (2010), we obtain that ξ_N is of order N^{1-av} when $v < 1/(a+1)$. Thus, Condition 3 holds provided that $v < 1/(2a+5)$. In particular, for the regression function in Example 1 of Wan et al. (2010), we have $a = 13/6$, so Condition 3 holds if $v < 3/28$. This example is based on the entire finite population. Note that the minimum risk ξ_N defined in this paper is influenced by the sampling design and may be inflated, since the parameters in each candidate model are estimated from the survey sample s rather than from the finite population U. Such inflation helps relax Condition 3. Further discussion of high-dimensional regression is provided in Supplementary Material S4.

Theorem 1 *If Conditions 1–3 hold, then*

$$E\left(C_n(\boldsymbol{\omega})\right) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o\left(R_N(\boldsymbol{\omega})\right).$$

Moreover, if $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ *and* $k_M^2 = O(N)$ *also hold, then*

$$E\left(C_n^*(\boldsymbol{\omega})\right) = R_N(\boldsymbol{\omega}) + N\sigma^2 + o\left(R_N(\boldsymbol{\omega})\right).$$

Proof. See Supplementary Material S3.2.

Since the remainder term $o(R_N(\boldsymbol{\omega}))$ is asymptotically negligible relative to $R_N(\boldsymbol{\omega})+N\sigma^2$, Theorem 1 establishes that $C_n(\boldsymbol{\omega})$ and $C_n^*(\boldsymbol{\omega})$ are asymptotically unbiased for $R_N(\boldsymbol{\omega})+N\sigma^2$. Furthermore, under the interchangeability condition $E_\epsilon E_d(\cdot) = E_d E_\epsilon(\cdot)$, Theorem 1 remains valid when Condition 3 is replaced by the weaker assumption $k_M(f\xi_N)^{-1} = o(1)$. In addition, based on the proof of Theorem 2 below, we can show that $C_n(\boldsymbol{\omega})$ and $C_n^*(\boldsymbol{\omega})$ are consistent estimators of $R_N(\boldsymbol{\omega}) + N\sigma^2$. See Corollaries 1–2 in Supplementary Material S2 for details.

Next, we establish the asymptotic optimality of the proposed model averaging estimator, which requires an additional regularity condition.

Condition 4 *For some fixed integer $1 \le G < \infty$, $E_\epsilon(e_i^{4G})$ is uniformly bounded in i, and $M\tilde{\xi}_N^{-2G}\sum_{m=1}^{M}(\tilde{R}_N(\boldsymbol{\omega}_m^0))^G = o(1)$, where $\boldsymbol{\omega}_m^0$ is an $M\times 1$ vector with 1 in the mth position and 0 elsewhere.*

Condition 4 is adopted from Wan et al. (2010), in which the validity of this condition is discussed in detail. This condition concerns only the superpopulation model and does not depend on the sampling design.

Theorem 2 *If Conditions 1–4 hold, then*

$$\frac{L_N(\hat{\boldsymbol{\omega}})}{\inf_{\boldsymbol{\omega}\in\mathcal{W}}\{L_N(\boldsymbol{\omega})\}} = 1 + o_p(1). \tag{12}$$

Moreover, if $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ and $k_M^2 = O(N)$ also hold, then

$$\frac{L_N(\hat{\boldsymbol{\omega}}^*)}{\inf_{\boldsymbol{\omega}\in\mathcal{W}}\{L_N(\boldsymbol{\omega})\}} = 1 + o_p(1). \tag{13}$$

Proof. See Supplementary Material S3.4.

Theorem 2 shows that the weight vector obtained by minimizing either $C_n(\boldsymbol{\omega})$ or $C_n^*(\boldsymbol{\omega})$ is asymptotically optimal in the sense of achieving the lowest possible loss.

Now, we establish the consistency of the weight estimators $\hat{\boldsymbol{\omega}}$ and $\hat{\boldsymbol{\omega}}^*$. Assume that the theoretically optimal weight vector $\boldsymbol{\omega}^*$ lies in the interior of $\mathcal{W}$. Define $\hat{\boldsymbol{\mu}}_{(m)s} = [\hat{\mu}_{i(m)}]_{i\in s}$

and $\boldsymbol{V}_1 = (\hat{\boldsymbol{\mu}}_{(1)s}, \ldots, \hat{\boldsymbol{\mu}}_{(M)s})$.

Condition 5 *$\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ and $c_7 \leq \lambda_{\min}(\boldsymbol{V}_1'\boldsymbol{\Phi}\boldsymbol{V}_1/N) \leq \lambda_{\max}(\boldsymbol{V}_1'\boldsymbol{\Phi}\boldsymbol{V}_1/N) \leq c_8$ in probability tending to 1, where c_7 and c_8 are positive constants.*

Condition 6 *$k_M(f\xi_N)^{-1} = o(1)$ and $M = O(N^{2\delta})$ for some $\delta > 0$.*

Condition 5 is a regularity assumption requiring the minimum and maximum eigenvalues of $\boldsymbol{V}_1'\boldsymbol{\Phi}\boldsymbol{V}_1/N$ to be bounded away from zero and infinity. Similar conditions can be found in Zhang et al. (2016) and Liao et al. (2019). Like Condition 3, Condition 6 restricts the growth rates of the number of candidate models and their dimensions. Its first restriction is weaker than Condition 3, while the second becomes weaker as δ increases. However, the range of δ is constrained by Theorem 3 below.

Theorem 3 *If Conditions 1–2 and 5–6 hold, then*

$$\|\hat{\boldsymbol{\omega}} - \boldsymbol{\omega}^*\| = O_p\left(\xi_N^{1/2} N^{-1/2+\delta}\right). \tag{14}$$

Moreover, if $k_M^2 = O(N)$ also holds, then

$$\|\hat{\boldsymbol{\omega}}^* - \boldsymbol{\omega}^*\| = O_p\left(\xi_N^{1/2} N^{-1/2+\delta}\right). \tag{15}$$

Proof. See Supplementary Material S3.6.

Theorem 3 shows that both $\hat{\boldsymbol{\omega}}$ and $\hat{\boldsymbol{\omega}}^*$ converge to $\boldsymbol{\omega}^*$ provided that $\xi_N = o(N^{1-2\delta})$. A smaller value of δ weakens this requirement and yields a faster convergence rate for the estimated weights. However, it also strengthens the second restriction in Condition 6, thereby revealing a trade-off. Under the setting considered in the discussion of Condition 3, Condition 6 and $\xi_N = o(N^{1-2\delta})$ hold if $v < 1/(a+1)$ and $v \leq 2\delta < av$. In particular, taking $v = 2\delta$ (so that $M/N^{2\delta}$ is bounded), they are satisfied if $v < 1/(a+1)$ and $a > 1$. Moreover, Theorem 3 implies that the more slowly ξ_N diverges, the faster the estimated weight vectors $\hat{\boldsymbol{\omega}}$ and $\hat{\boldsymbol{\omega}}^*$ converge to the theoretically optimal weight vector $\boldsymbol{\omega}^*$.

In the following, we consider a scenario in which the candidate model set contains one or more correct models (i.e., models that include the true model), and show that our proposed method asymptotically allocates all the weights to these correct models. Suppose that there are s_0 $(1 \leq s_0 \leq M)$ correct models in the candidate models. For simplicity, we denote $M_1, \ldots, M_{s_0}$ as the correct models and $M_{s_0+1}, \ldots, M_M$ as the misspecified models. Let $\mathbf{\Omega}^*$ be the subset of $\{1, \ldots, M\}$ that indexes all correct models. For the correct model set $\mathbf{\Omega}^*$, define the corresponding weight space in $\mathbb{R}^M$ as $\mathcal{W}_1 = \{\boldsymbol{\omega} \in \mathcal{W} : \sum_{m=1}^{s_0} \omega_m = 1\}$. Let $\mathcal{W}_2 = \mathcal{W} - \mathcal{W}_1$ and $\tilde{\xi}_N^* = \inf_{\boldsymbol{\omega} \in \mathcal{W}_2} \{\tilde{R}_N(\boldsymbol{\omega})\}$. Within this framework, Conditions 3 and 4 are replaced by the following assumptions to obtain the desired result.

Condition 7 $M^2\sqrt{nk_M}(f\tilde{\xi}_N^*)^{-1} = o(1)$.

Condition 8 *For some fixed integer* $1 \leq G < \infty$, $E_\epsilon\left(e_i^{4G}\right)$ *is bounded uniformly in* i, *and* $M\tilde{\xi}_N^{*-2G}\sum_{m=1}^{M}(\tilde{R}_N(\boldsymbol{\omega}_m^0))^G = o(1)$.

Note that $\tilde{\xi}_N^*$ represents the minimum risk over the weight space $\mathcal{W}_2 \subset \mathcal{W}$. Therefore, Conditions 7 and 8 are milder than Conditions 3 and 4, respectively.

Theorem 4 *If Conditions 1–2 and 7–8 hold, then*

$$P\left(\hat{\boldsymbol{\omega}} \in \mathcal{W}_1\right) \to 1. \tag{16}$$

Moreover, if $\boldsymbol{\mu}_U'\boldsymbol{\mu}_U = O(N)$ *and* $k_M^2 = O(N)$ *also hold, then*

$$P\left(\hat{\boldsymbol{\omega}}^* \in \mathcal{W}_1\right) \to 1. \tag{17}$$

Proof. See Supplementary Material S3.7.

Theorem 4 shows that the weights obtained by minimizing either the criterion $C_n(\boldsymbol{\omega})$ or $C_n^*(\boldsymbol{\omega})$ asymptotically concentrate on the correct models.

4 Model-assisted averaging estimation for finite population

In this section, we develop a model-assisted averaging estimator for the population total under the superpopulation model (3) and establish its theoretical properties.

4.1 Model-assisted averaging estimator of the population total

As described in Section 2, we draw a sample s of size n from the finite population U according to a sampling design $P(s)$. The first- and second-order inclusion probabilities are denoted by π_i and π_{ij}, respectively. Suppose that $\{(y_i, \boldsymbol{x}_i'), i \in U\}$ are generated from the superpopulation model (3). Our objective is to estimate the population total $t_y = \sum_U y_i$, assuming that we have observed $(y_i, \boldsymbol{x}_i')$ for $i \in s$, and $\boldsymbol{t}_x = \sum_U \boldsymbol{x}_i$, the population total of auxiliary vector, is also known.

In Section 3.1, based on the SMA estimator $\hat{\mu}_i(\boldsymbol{\omega})$, we construct the model-assisted averaging estimator of t_y given in Equation (7). However, the weight vector $\boldsymbol{\omega}$ is unknown. We therefore estimate it using $\hat{\boldsymbol{\omega}}$, as defined in Equation (9) of Section 3.2, which yields the following model-assisted averaging estimator:

$$\hat{t}_y(\hat{\boldsymbol{\omega}}) = \sum_{i \in s} \frac{y_i - \hat{\mu}_i(\hat{\boldsymbol{\omega}})}{\pi_i} + \sum_{m=1}^{M} \hat{\omega}_m \left(\boldsymbol{t}_{x(m)}' \hat{\boldsymbol{\beta}}_{(m)} \right). \tag{18}$$

When σ^2 is unknown, we estimate $\boldsymbol{\omega}$ using the feasible estimator $\hat{\boldsymbol{\omega}}^*$ defined in Equation (11). The resulting model-assisted averaging estimator of t_y is

$$\hat{t}_y(\hat{\boldsymbol{\omega}}^*) = \sum_{i \in s} \frac{y_i - \hat{\mu}_i(\hat{\boldsymbol{\omega}}^*)}{\pi_i} + \sum_{m=1}^{M} \hat{\omega}_m^* \left(\boldsymbol{t}_{x(m)}' \hat{\boldsymbol{\beta}}_{(m)} \right). \tag{19}$$

4.2 Theoretical properties

In this subsection, we establish several theoretical properties of the proposed model-assisted averaging estimators. For ease of presentation, let $\bar{t}_y = t_y / N$ denote the finite population

mean and define its model-assisted averaging estimator as $\hat{\bar{t}}_y(\boldsymbol{\omega}) = \hat{t}_y(\boldsymbol{\omega})/N$. Our subsequent analysis focuses on the theoretical properties of estimators of the finite population mean. Let $\hat{\boldsymbol{\omega}}_0$ be an arbitrary estimator of the optimal weight vector $\boldsymbol{\omega}^*$, and define $\zeta_n = E\left\|\hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^*\right\|^4$. Define a difference estimator of the finite population mean $\bar{t}_y$ as follows:

$$\hat{\bar{t}}_y = \frac{1}{N}\left(\sum_{i\in s}\frac{y_i-\mu_i}{\pi_i} + \sum_{i\in U}\mu_i\right).$$

Its mean squared error (MSE) under the joint model–design distribution is

$$\mathrm{MSE}\left(\hat{\bar{t}}_y\right) = \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j}E_\epsilon\left\{\left(y_i-\mu_i\right)\left(y_j-\mu_j\right)\right\}.$$

Condition 9 *Some assumptions for higher-order inclusion probabilities:*

$$\limsup_{n\to\infty} n \max_{(i,j,k)\in\boldsymbol{D}_3}\left|E_d\{(I_i-\pi_i)^2(I_j-\pi_j)(I_k-\pi_k)\}\right| < \infty,$$

$$\limsup_{n\to\infty} \max_{(i,j,k,l)\in\boldsymbol{D}_4}\left|E_d\{(I_iI_j-\pi_{ij})(I_kI_l-\pi_{kl})\}\right| = 0,$$

and

$$\limsup_{n\to\infty} n^2 \max_{(i,j,k,l)\in\boldsymbol{D}_4}\left|E_d\{(I_i-\pi_i)(I_j-\pi_j)(I_k-\pi_k)(I_l-\pi_l)\}\right| < \infty,$$

where $\boldsymbol{D}_t$ *is the set of all distinct* t*-tuples* $(i_1, i_2, \ldots, i_t)$ *from* U*.*

Condition 10 $E_\epsilon(e_i^4)$ *is uniformly bounded in* i*, and* $\sum_U \mu_i^4 = O(N)$*.*

Condition 11 $\max\{k_M^2, Mk_M\} = o(n)$ *and* $\zeta_n^{1/2}Mk_M^2 = o(1)$*.*

Condition 9 extends Condition 2, a standard assumption on higher-order inclusion probabilities. As shown in Breidt & Opsomer (2000) and Boistard et al. (2012), this condition holds for simple random sampling without replacement, Poisson sampling, and rejective sampling, but generally fails for designs with nontrivial clustering. Conditions 2 and 9 have been widely used to establish the consistency and asymptotic normality of complex survey estimators (Breidt et al. 2007, Wang 2009, Cardot et al. 2010, Wang & Wang 2011, Zong et al. 2026). Condition 10 imposes mild moment restrictions on e_i and μ_i.

Condition 11 constrains the growth rates of the maximum model dimension and the number of candidate models. When M and k_M are bounded, this condition holds provided that $\zeta_n = o(1)$. It can also be satisfied when M and k_M diverge. To illustrate this, let $\hat{\boldsymbol{\omega}}_0$ be either $\hat{\boldsymbol{\omega}}$ or $\hat{\boldsymbol{\omega}}^*$. Motivated by Theorem 3, it is reasonable to assume that $\zeta_n^{1/2}$ is of order $\xi_N N^{-1+2\delta}$. Under the setting considered in the discussion of Condition 3, Conditions 6 and 11 hold if $v < 1/(a+1)$ and $v \leq 2\delta < (a-3)v$. In particular, taking $v = 2\delta$, Conditions 6 and 11 are satisfied when $v < 1/(a+1)$ and $a > 4$.

Theorem 5 *If Conditions 1–2 and 9–10 hold, then*

$$\mathrm{MSE}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) = \mathrm{MSE}\left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*)\right) + O\left(\frac{1}{n}\sqrt{\zeta_n^{1/2} M k_M^2 \left(1 + \frac{k_M}{\sqrt{n}}\right)}\right),$$

where

$$\mathrm{MSE}\left(\hat{\bar{t}}_y(\boldsymbol{\omega}^*)\right) = \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j} E_\epsilon\left\{\left(y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right)\left(y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right)\right\} + O\left(\frac{k_M}{n^{3/2}}\right).$$

Further, if Condition 11 holds, then

$$\mathrm{MSE}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) = \frac{1}{N^2}\sum_{i,j\in U}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_i\pi_j} E_\epsilon\left\{\left(y_i - \tilde{\mu}_i(\boldsymbol{\omega}^*)\right)\left(y_j - \tilde{\mu}_j(\boldsymbol{\omega}^*)\right)\right\} + o\left(n^{-1}\right).$$

Proof. See Supplementary Material S3.9.

Theorem 5 provides the MSE of $\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)$ under certain regularity conditions. When Condition 11 holds, the MSEs of $\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)$ and $\hat{\bar{t}}_y(\boldsymbol{\omega}^*)$ are asymptotically equivalent. The next theorem presents an asymptotically unbiased and consistent estimator of $\mathrm{MSE}(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0))$.

Theorem 6 *If Conditions 1–2 and 9–11 hold, then*

$$\lim_{n\to\infty} nE\left|\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) - \mathrm{MSE}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)\right| = o(1),$$

where

$$\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right) = \frac{1}{N^2}\sum_{i,j\in s}\frac{\pi_{ij}-\pi_i\pi_j}{\pi_{ij}\pi_i\pi_j}\left\{y_i - \hat{\mu}_i(\hat{\boldsymbol{\omega}}_0)\right\}\left\{y_j - \hat{\mu}_j(\hat{\boldsymbol{\omega}}_0)\right\}.$$

Proof. See Supplementary Material S3.10.

Based on Theorem 6 and Markov's inequality, $\widehat{\mathrm{MSE}}(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0))$ is an asymptotically unbiased and consistent estimator of $\mathrm{MSE}(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0))$. Now, we establish the asymptotic normality of the model-assisted averaging estimator $\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)$.

Condition 12 *$\tilde{R}_N(\boldsymbol{\omega}^*) = o(N)$ and $n\mathrm{MSE}(\hat{\bar{t}}_{y,\mathrm{diff}}) > c_9$, where c_9 is a positive constant.*

The first part of Condition 12 is a regularity assumption on the risk function. In particular, when the candidate set contains correct models, we can show that $\tilde{R}_N(\boldsymbol{\omega}_0) = o(N)$ for any $\boldsymbol{\omega}_0 \in \mathcal{W}_1$. Since $\boldsymbol{\omega}^*$ attains the minimum risk ξ_N, it is likely to belong to $\mathcal{W}_1$. So assuming $\tilde{R}_N(\boldsymbol{\omega}^*) = o(N)$ is reasonable. Since the MSE of $\hat{\bar{t}}_y$ is positive and of order $O(n^{-1})$ (Särndal et al. 1992), the second part of Condition 12 is also a mild and standard assumption.

Theorem 7 *If Conditions 1–2 and 9–12 hold, then*

$$\frac{\hat{\bar{t}}_y - \bar{t}_y}{\sqrt{\mathrm{MSE}\left(\hat{\bar{t}}_y\right)}} \xrightarrow{d} N(0,1)$$

implies

$$\frac{\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y}{\sqrt{\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}} \xrightarrow{d} N(0,1).$$

Proof. See Supplementary Material S3.11.

From Theorem 7, it follows that establishing a central limit theorem (CLT) for the model-assisted averaging estimator reduces to establishing a CLT for the difference estimator, which is a problem similar to establishing a CLT for the classical HT estimator.

Under the design-based inference framework, the design properties of model-assisted estimators are often investigated. Hájek (1960) and Hájek (1964) proved that the HT estimator is asymptotically design normal under simple random sampling without replacement and rejective sampling, respectively, provided certain regularity conditions hold. As shown by Hájek (1959), the rejective sampling design $R(s)$ maximizes entropy within the class of

all fixed-size sampling designs without replacement. Furthermore, Berger (1996) established the asymptotic design normality of the HT estimator when the sampling design exhibits a small divergence from rejective sampling. In a similar vein, we establish the asymptotic design normality of the proposed model-assisted averaging estimator $\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)$.

Denote the measure of divergence between a sampling design $P(s)$ and the rejective sampling design $R(s)$ by

$$D(P\|R) = \sum_{s\in\Psi} P(s)\log\left\{\frac{P(s)}{R(s)}\right\},$$

where Ψ denotes the set of all samples of n distinct units drawn from U. For the rejective sampling design $R(s)$, define $d_R = \sum_{i\in U}\{\pi_i(1-\pi_i)\}$, $g_R = d_R^{-1}\sum_{i\in U}\{y_i(1-\pi_i)\}$, $v_i = y_i - \pi_i g_R$, and $\sigma_R^2 = \sum_{i\in U}\{v_i^2\pi_i^{-1}(1-\pi_i)\}$. Let $e_R = \inf\{\varepsilon : L(\varepsilon) \le \varepsilon\}$, where

$$L(\varepsilon) = \sum_{\{i\in U:|v_i|>\varepsilon\pi_i\sigma_R\}} \frac{(1-\pi_i)v_i^2}{\pi_i\sigma_R^2}.$$

Condition 13 *$|y_i|$ and $|\tilde{\mu}_{i(m)}|$ are uniformly bounded in i and m.*

Condition 14 *$M^2k_M^2 = o(n)$ and $\tilde{\zeta}_n^{1/2}M = o(1)$, where $\tilde{\zeta}_n = E_d\|\hat{\boldsymbol{\omega}}_0 - \boldsymbol{\omega}^*\|^4$.*

Condition 15 *$\tilde{L}_N(\boldsymbol{\omega}^*) = o(N)$ and $n\mathrm{MSE}_d(\hat{\bar{t}}_{y,\mathrm{diff}}) > c_{10}$, where c_{10} is a positive constant.*

Condition 13 is commonly used in the design-based inference framework (Breidt et al. 2007, Zong et al. 2019). Conditions 14 and 15 extend Conditions 11 and 12, respectively, which do not consider the model randomness.

Theorem 8 *Assume that for the sampling design $P(s)$, $D(P\|R) \to 0$, $e_R \to 0$, and $d_R \to \infty$. If Conditions 1–2 and 13–15 hold, then*

$$\frac{\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0) - \bar{t}_y}{\sqrt{\widehat{\mathrm{MSE}}\left(\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)\right)}} \xrightarrow{\tilde{d}} N(0,1),$$

where $\tilde{d}$ denotes convergence in distribution with respect to the sampling design.

Proof. See Supplementary Material S3.13.

It follows from Theorem 8 and Berger (1996) that, for several common sampling designs, such as simple random sampling without replacement, rejective sampling, and Rao-Sampford sampling, the proposed model-assisted averaging estimator $\hat{\bar{t}}_y(\hat{\boldsymbol{\omega}}_0)$ is asymptotically design normal.

5 Numerical analyses

In this section, we assess the performance of the proposed method through simulation studies and empirical applications.

5.1 Simulation studies

Following Hansen (2007), we generate a finite population U of size N according to

$$y_i = \sum_{j=1}^{M_0} \beta_j x_{ij} + e_i, \quad i = 1, \ldots, N, \tag{20}$$

where $\beta_j = c\sqrt{2\alpha} j^{-\alpha-1/2}$, $x_{i1} = 1$, and the remaining covariates $\{x_{ij}\}$ and the errors $\{e_i\}$ are independent and identically distributed (i.i.d.) $N(0,1)$. In this setting, $R^2 = c^2/(1+c^2)$ is determined by the parameter c, and α controls the rate of decay of the regression coefficients. An additional auxiliary variable is generated as $z_i = \rho y_i/\{(1-\rho^2)v_y\}^{1/2} + \epsilon_i$, where v_y is the finite population variance of $\{y_i\}$, ρ controls the correlation between y_i and z_i, and $\{\epsilon_i\}$ are i.i.d. $N(10,1)$. To implement πps sampling, we set $\pi_i \propto z_i$ for all $i \in U$ and draw a sample using the Sunter method (see the R package *TeachingSampling*).

We compare the following model selection and averaging methods: (a) AIC model selection (AIC). For the mth candidate model, $\mathrm{AIC}_m = n \log \hat{Q}_m^2 + 2k_m$, where $\hat{Q}_m^2$ is the residual sum of squares computed using the design-weighted estimator $\hat{\boldsymbol{\beta}}_{(m)}$. (b) BIC model selection (BIC), with $\mathrm{BIC}_m = n \log \hat{Q}_m^2 + k_m \log n$. (c) Mallows C_p model selection (C_p),

with $C_{p,m} = (n - k_M)\hat{Q}_m^2/\hat{Q}_M^2 + 2k_m$. (d) Adaptive LASSO model selection (ALASSO), with the penalty $\lambda \sum_{j=1}^{k_M} |\hat{\beta}_{0j}|^{-\gamma}|\beta_j|$, where $\gamma = 1$ and $\hat{\beta}_{0j}$ is the jth component of the least squares estimator under the largest candidate model (Zou 2006). (e) SCAD model selection (SCAD), with the penalty proposed by Fan & Li (2001). For both ALASSO and SCAD, the selected model consists of the variables with nonzero estimated coefficients, and the tuning parameter λ is determined by ten-fold cross-validation. (f) Smoothed AIC model averaging (SAIC), where $\omega_m = \exp\left(-\frac{1}{2}\widetilde{\text{AIC}}_m\right) / \sum_{j=1}^{M} \exp\left(-\frac{1}{2}\widetilde{\text{AIC}}_j\right)$ and $\widetilde{\text{AIC}}_m = \text{AIC}_m - \min_{j=1,\ldots,M}\{\text{AIC}_j\}$. (g) Smoothed BIC model averaging (SBIC), where $\omega_m = \exp\left(-\frac{1}{2}\widetilde{\text{BIC}}_m\right) / \sum_{j=1}^{M} \exp\left(-\frac{1}{2}\widetilde{\text{BIC}}_j\right)$ and $\widetilde{\text{BIC}}_m = \text{BIC}_m - \min_{j=1,\ldots,M}\{\text{BIC}_j\}$. (h) Convex aggregation based on empirical risk minimization (AGG1), following the procedure described in Section 5.2.2 of Nemirovski (2000). (i) Penalized least squares aggregation (AGG2), with the ℓ_1-type penalty given in Equation (2.4) of Bunea et al. (2007). For the aggregation methods AGG1 and AGG2, the sample is randomly split into a training set and a validation set of equal size. (j) The proposed SMA method (SMA1), with the penalty $2\hat{\sigma}_{(M)}^2 k(\boldsymbol{\omega})$. (k) The proposed SMA method (SMA2), with the penalty $2\hat{\sigma}_{(M)}^2 k_s(\boldsymbol{\omega})$.

Similar to Hansen (2007), we consider the nested candidate models, with the mth model containing the first m regressors, and set the number of candidate models to either $M = [3n^{1/3}]$ or $M = [n/3]$. We set $N = 2000$ and $M_0 = 500$. The R^2 is varied over $\{0.1, 0.2, \ldots, 0.9\}$, while $\alpha \in \{0.5, 1.0\}$ and $\rho \in \{0.1,\ 0.6\}$. We view model selection as a special case of model averaging with 0–1 weights. For each method considered in the comparison, let $\check{\boldsymbol{\omega}}$ denote the corresponding weight estimator. We evaluate the resulting estimators using the following two performance measures:

$$\text{MSE}_{\mu} = \frac{1}{D}\sum_{d=1}^{D}\left\{\frac{1}{N}\sum_{i\in U}\left(\hat{\mu}_i^{(d)}(\check{\boldsymbol{\omega}}) - \mu_i^{(d)}\right)^2\right\}$$

and

$$\text{MSE}_{\bar{y}} = \frac{1}{D}\sum_{d=1}^{D}\left(\hat{\bar{t}}_y^{(d)}(\check{\boldsymbol{\omega}}) - \bar{t}_y^{(d)}\right)^2,$$

where $D = 1000$, and $\hat{\mu}_i^{(d)}(\check{\boldsymbol{\omega}})$ and $\hat{\bar{t}}_y^{(d)}(\check{\boldsymbol{\omega}})$ are the averaging estimators of $\mu_i^{(d)}$ and $\bar{t}_y^{(d)}$ in the dth replication, respectively.

Table 1 reports the values of MSE_μ for the different methods when $M = [3n^{1/3}]$ and $n = 100$. Results for the other sample sizes ($n = 50, 150, 200$) are presented in Tables S1–S3 of Supplementary Material S5.1. The smallest and second-smallest values in each row (excluding SMA2) are highlighted in bold and italics, respectively. Overall, the proposed methods (SMA1 and SMA2) perform best and yield similar results. The model averaging methods (SAIC, SBIC, and SMA1) generally outperform their model selection counterparts (AIC, BIC, and C_p). SBIC performs well when R^2 is small, whereas SAIC gradually improves as the sample size increases. Tables S7–S10 in Supplementary Material S5.2 show the simulation results for $M = [n/3]$. Consistent with the preceding findings, the proposed methods still perform best in most settings. SBIC is competitive when α is large and R^2 is low. Its stronger penalty may help control model complexity when the candidate models are high-dimensional and the true regression coefficients decay rapidly.

Table 2 reports the values of $\text{MSE}_{\bar{y}} \times 100$ for the different methods when $M = [3n^{1/3}]$ and $n = 100$. Results for the other sample sizes ($n = 50, 150, 200$) are presented in Tables S4–S6 of Supplementary Material S5.1. The proposed methods (SMA1 and SMA2) yield similar results and perform best in most settings. The model averaging methods (SAIC, SBIC, and SMA1) generally outperform their model selection counterparts (AIC, BIC, and C_p). Tables S11–14 in Supplementary Material S5.2 present the simulation results for $M = [n/3]$. The results show that the proposed methods exhibit good performance when R^2 is high. AGG2 performs well when ρ and α are large, possibly because its ℓ_1-type penalty is beneficial when the number of covariates is non-negligible relative to the sample size.

We also report simulation results using the true variance $\sigma^2 = 1$ in Supplementary Material S5.3. These results are largely consistent with those obtained using an estimated

Table 1: Values of MSE_μ for various methods in the simulation ($n = 100$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
	0.1	0.1218	0.1391	0.0735	0.1813	0.0960	0.0756	0.0812	**0.0672**	0.0935	*0.0690*	0.0689
	0.2	0.1742	0.1736	*0.1095*	0.3736	0.1353	0.1097	0.1362	0.1148	0.1336	**0.1014**	0.1014
	0.3	0.2055	0.2016	*0.1399*	0.4498	0.1689	0.1405	0.1814	0.1549	0.1676	**0.1299**	0.1298
$\rho = 0.1$	0.4	0.2317	0.2327	0.1787	0.4290	0.2044	*0.1749*	0.2314	0.1999	0.2024	**0.1638**	0.1639
$\alpha = 0.5$	0.5	0.2544	0.2631	0.2262	0.4081	0.2412	*0.2125*	0.2823	0.2494	0.2418	**0.2013**	0.2013
	0.6	0.2923	0.3056	0.2814	0.4105	0.2935	*0.2658*	0.3510	0.3124	0.2959	**0.2540**	0.2541
	0.7	0.3619	0.3769	0.3758	0.4816	0.3708	*0.3452*	0.4580	0.4083	0.3740	**0.3344**	0.3346
	0.8	0.4802	0.5046	0.5294	0.6060	0.4971	*0.4784*	0.6128	0.5589	0.5031	**0.4695**	0.4697
	0.9	**0.8668**	0.8889	0.9828	1.0546	0.8857	0.8778	1.0377	0.9778	0.8930	*0.8769*	0.8772
	0.1	0.0953	0.1168	0.0591	0.2585	0.0721	0.0561	0.0542	**0.0426**	0.0692	*0.0510*	0.0507
	0.2	0.1270	0.1333	0.0737	0.3720	0.0867	0.0691	0.0789	**0.0632**	0.0841	*0.0639*	0.0638
	0.3	0.1428	0.1395	0.0868	0.3132	0.0937	*0.0769*	0.0955	0.0780	0.0918	**0.0733**	0.0732
$\rho = 0.1$	0.4	0.1536	0.1535	0.0998	0.2395	0.1117	0.0913	0.1080	*0.0897*	0.1089	**0.0848**	0.0847
$\alpha = 1.0$	0.5	0.1639	0.1619	0.1140	0.2186	0.1255	*0.1025*	0.1261	0.1043	0.1226	**0.0956**	0.0955
	0.6	0.1726	0.1706	0.1324	0.2104	0.1359	*0.1128*	0.1463	0.1243	0.1334	**0.1088**	0.1087
	0.7	0.1819	0.1824	0.1490	0.2006	0.1535	*0.1289*	0.1703	0.1448	0.1514	**0.1225**	0.1225
	0.8	0.1924	0.1965	0.1799	0.2211	0.1761	*0.1510*	0.2023	0.1751	0.1755	**0.1456**	0.1456
	0.9	0.2239	0.2349	0.2499	0.2823	0.2241	*0.2016*	0.2731	0.2386	0.2253	**0.1959**	0.1960
	0.1	0.1252	0.1428	0.0740	0.1812	0.0995	0.0792	0.0819	**0.0672**	0.0971	*0.0707*	0.0704
	0.2	0.1752	0.1760	*0.1069*	0.3525	0.1355	0.1114	0.1379	0.1154	0.1328	**0.1025**	0.1023
	0.3	0.2102	0.2046	*0.1394*	0.3974	0.1693	0.1410	0.1817	0.1566	0.1673	**0.1306**	0.1305
$\rho = 0.6$	0.4	0.2268	0.2278	0.1735	0.3695	0.2023	*0.1724*	0.2299	0.2000	0.2011	**0.1619**	0.1619
$\alpha = 0.5$	0.5	0.2543	0.2605	0.2215	0.3586	0.2442	*0.2145*	0.2855	0.2513	0.2445	**0.2039**	0.2040
	0.6	0.2960	0.3083	0.2820	0.3851	0.2947	*0.2668*	0.3588	0.3193	0.2962	**0.2560**	0.2562
	0.7	0.3606	0.3792	0.3716	0.4460	0.3723	*0.3455*	0.4540	0.4096	0.3764	**0.3348**	0.3350
	0.8	0.4794	0.5021	0.5270	0.5899	0.4983	*0.4787*	0.6100	0.5541	0.5032	**0.4698**	0.4700
	0.9	**0.8683**	0.8937	0.9693	1.0356	0.8956	0.8823	1.0311	0.9729	0.9029	*0.8772*	0.8776
	0.1	0.0970	0.1184	0.0577	0.2522	0.0709	0.0558	0.0546	**0.0431**	0.0683	*0.0518*	0.0514
	0.2	0.1251	0.1327	0.0709	0.3300	0.0844	0.0672	0.0793	**0.0630**	0.0818	*0.0634*	0.0632
	0.3	0.1445	0.1437	0.0809	0.2568	0.0952	*0.0762*	0.0921	0.0772	0.0941	**0.0732**	0.0730
$\rho = 0.6$	0.4	0.1560	0.1554	0.0974	0.2071	0.1128	0.0929	0.1092	*0.0918*	0.1111	**0.0872**	0.0870
$\alpha = 1.0$	0.5	0.1629	0.1614	0.1092	0.1880	0.1261	*0.1031*	0.1264	0.1053	0.1231	**0.0966**	0.0965
	0.6	0.1716	0.1709	0.1240	0.1830	0.1371	*0.1141*	0.1458	0.1230	0.1349	**0.1080**	0.1079
	0.7	0.1822	0.1823	0.1454	0.1939	0.1528	*0.1298*	0.1702	0.1449	0.1515	**0.1235**	0.1235
	0.8	0.2009	0.2045	0.1809	0.2148	0.1820	*0.1570*	0.2075	0.1793	0.1816	**0.1507**	0.1507
	0.9	0.2256	0.2365	0.2440	0.2749	0.2243	*0.2015*	0.2731	0.2396	0.2254	**0.1969**	0.1970

Table 2: Values of $\text{MSE}_{\bar{y}} \times 100$ for various methods in the simulation ($n = 100$)

	R^2	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
	0.1	0.9986	1.0217	0.9553	0.9654	0.9663	0.9554	0.9702	**0.9503**	0.9607	*0.9544*	0.9543
	0.2	1.2140	1.2234	*1.1358*	1.1994	1.1628	1.1378	1.1857	1.1551	1.1667	**1.1341**	1.1342
	0.3	1.1274	1.1222	*1.0608*	1.1089	1.0944	1.0692	1.1059	1.0779	1.0946	**1.0569**	1.0569
$\rho = 0.1$	0.4	1.1256	1.1289	1.0827	1.1563	1.1046	*1.0790*	1.1278	1.0965	1.1077	**1.0593**	1.0592
$\alpha = 0.5$	0.5	1.1476	1.1492	*1.1090*	1.1680	1.1364	1.1096	1.1809	1.1505	1.1347	**1.1005**	1.1004
	0.6	1.1515	1.1706	1.1522	1.2039	1.1603	*1.1498*	1.2982	1.2471	1.1813	**1.1488**	1.1494
	0.7	*1.3318*	1.3567	1.3405	1.3680	1.3628	1.3360	1.4654	1.4078	1.3667	**1.3186**	1.3189
	0.8	1.3742	1.3936	1.3972	1.4126	1.3945	*1.3705*	1.4758	1.4270	1.4027	**1.3659**	1.3657
	0.9	**1.8197**	1.8419	1.9761	1.9840	*1.8242*	1.8370	2.0952	1.9988	1.8337	1.8466	1.8470
	0.1	1.0908	1.1189	1.0463	**1.0337**	1.0778	1.0540	1.0369	*1.0369*	1.0758	1.0496	1.0494
	0.2	1.1143	1.1272	1.0583	1.0718	1.0617	**1.0431**	1.0657	*1.0453*	1.0569	1.0459	1.0458
	0.3	1.0798	1.0729	**1.0049**	1.0358	1.0201	1.0114	1.0163	1.0101	1.0208	*1.0100*	1.0101
$\rho = 0.1$	0.4	1.0209	1.0335	1.0020	1.0033	1.0122	*0.9960*	1.0301	1.0150	1.0105	**0.9921**	0.9921
$\alpha = 1.0$	0.5	1.1507	1.1318	1.0867	1.0868	1.1001	**1.0855**	1.1274	1.0997	1.1014	*1.0861*	1.0863
	0.6	1.1292	1.1240	1.0554	1.0558	1.0694	**1.0438**	1.0880	1.0630	1.0736	*1.0456*	1.0459
	0.7	1.0777	1.0896	1.0580	1.0538	1.0712	*1.0491*	1.0881	1.0710	1.0722	**1.0400**	1.0401
	0.8	1.1043	1.1049	1.1037	1.1144	1.0951	*1.0769*	1.1345	1.1166	1.1029	**1.0744**	1.0748
	0.9	1.1496	1.1643	1.1978	1.1919	1.1630	*1.1388*	1.2215	1.1772	1.1658	**1.1374**	1.1373
	0.1	1.0553	1.0899	*0.9969*	**0.9577**	1.0255	1.0112	1.0131	1.0000	1.0250	0.9989	0.9992
	0.2	1.1106	1.1232	1.0447	**1.0094**	1.0642	1.0517	1.0580	*1.0410*	1.0649	1.0418	1.0418
	0.3	1.1941	1.1710	*1.0976*	1.1452	1.1302	1.1024	1.1389	1.1117	1.1368	**1.0870**	1.0875
$\rho = 0.6$	0.4	1.1853	1.1945	*1.1386*	1.1559	1.1815	1.1508	1.2090	1.1763	1.1838	**1.1368**	1.1372
$\alpha = 0.5$	0.5	1.2333	1.2233	*1.1573*	**1.1508**	1.2308	1.1975	1.2275	1.1927	1.2381	1.1694	1.1696
	0.6	1.2236	1.2356	1.1879	1.1905	1.2090	*1.1856*	1.2913	1.2472	1.2005	**1.1769**	1.1769
	0.7	1.2817	1.3046	1.2553	*1.2531*	1.2829	1.2631	1.3708	1.3145	1.2815	**1.2453**	1.2455
	0.8	*1.3367*	1.3561	1.3535	1.3662	1.3641	1.3371	1.4328	1.3903	1.3558	**1.3164**	1.3166
	0.9	**1.8617**	1.9000	1.9206	1.9524	1.8959	1.8841	2.0379	1.9695	1.9020	*1.8719*	1.8721
	0.1	1.0821	1.1274	1.0363	**0.9714**	1.0568	1.0441	1.0321	*1.0235*	1.0577	1.0387	1.0390
	0.2	1.0961	1.0993	1.0398	**1.0005**	1.0606	1.0464	1.0619	1.0449	1.0622	*1.0398*	1.0400
	0.3	1.0955	1.0889	1.0219	**0.9901**	1.0461	1.0180	1.0373	1.0197	1.0486	*1.0179*	1.0181
$\rho = 0.6$	0.4	1.1260	1.1265	1.0642	**1.0326**	1.0883	1.0635	1.0752	1.0642	1.0829	*1.0550*	1.0553
$\alpha = 1.0$	0.5	1.1482	1.1420	*1.1097*	**1.0970**	1.1344	1.1157	1.1588	1.1309	1.1342	1.1166	1.1169
	0.6	1.0939	1.1004	*1.0366*	**1.0162**	1.0608	1.0425	1.0815	1.0609	1.0510	1.0402	1.0402
	0.7	1.1280	1.1295	1.0846	*1.0803*	1.1052	1.0835	1.1255	1.0921	1.1117	**1.0747**	1.0750
	0.8	1.2270	1.2376	1.1726	**1.1578**	1.1925	*1.1704*	1.2177	1.1910	1.1839	1.1737	1.1736
	0.9	1.0622	1.0736	1.1328	1.1310	1.0638	**1.0459**	1.1414	1.1102	1.0653	*1.0589*	1.0591

variance, with the proposed methods performing best overall, particularly in terms of MSE_μ. In Supplementary Material S5.4, we compare the methods under varying degrees of model misspecification. The results show that the proposed method outperforms the other methods overall. When the candidate models are severely underfitted ($M = 11$), the largest-model estimator performs well because it incorporates the most information. As M increases, the proposed method improves and achieves the best performance among the methods considered. When the largest candidate model is overfitted ($M > 51$), its performance gradually deteriorates as M increases, possibly because including irrelevant covariates increases the estimation variance. However, the proposed method still performs best. Moreover, Figure S1 in Supplementary Material S5.5 shows the sum of the estimated weights assigned by SMA1 to the correct models across different sample sizes. This sum approaches one as the sample size increases, numerically verifying the selection consistency of the weights established in Theorem 4. Convergence becomes faster as R^2 increases.

5.2 Empirical application 1

In this subsection, we compare the various methods based on the Beijing PM2.5 dataset (Liang et al. 2016). This dataset records hourly environmental observations from March 5, 2013 to December 31, 2015. The indicators we are concerned about include PM2.5 (PM), accumulated wind speed (Iws), humidity (HUMI), atmospheric pressure (PRES) and temperature (TEMP). The records with NA values are deleted, and the daily averages of five indicators are calculated. We take the logarithm for PM and Iws indicators, because they are skewed and greater than 0. Through the above processing, we obtain $N = 924$ daily average environmental observations, which is regarded as a finite population U. The indicators included in the ith observation are $y_i = \log(\text{PM})$, $x_{i1} = \log(\text{Iws})$, $x_{i2} = \text{HUMI}$, $x_{i3} = \text{TEMP}$, and $x_{i4} = \text{PRES}$.

To implement πPS sampling, we set the first-order inclusion probabilities to satisfy $\pi_i \propto x_{i1}$ for all $i \in U$. As in Subsection 5.1, we compare methods (a)–(k). Since the true mean μ_i is unknown for real data, the first performance measure are redefined as

$$\mathrm{MSE}_y = \frac{1}{D}\sum_{d=1}^{D}\left\{\frac{1}{N-n}\sum_{i\in U/s}\left(\hat{\mu}_i^{(d)}(\breve{\boldsymbol{\omega}}) - y_i^{(d)}\right)^2\right\}.$$

Let n be varied among $\{30, 50, 80, 100, 120, 150, 180, 200\}$. We consider the nested model framework, with a total of 5 candidate models (including intercept term). Repeatedly calculating D =1000 times, Tables 3 and 4 report the results for the different methods across various sample sizes. It can be found that the proposed methods (SMA1 and SMA2) perform best overall and yield similar results. The accuracy of all estimators improves as the sample size increases, and the model averaging methods (SAIC, SBIC, and SMA1) generally outperform their model selection counterparts (AIC, BIC, and C_p).

Table 3: Values of MSE_y for various methods in Empirical Application 1

n	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
30	0.6438	0.6439	0.6736	1.0231	0.6440	**0.6189**	0.6587	0.6290	0.6454	*0.6202*	0.6231
50	0.5865	0.5913	0.6126	0.8580	0.5927	*0.5765*	0.6100	0.5910	0.5936	**0.5745**	0.5777
80	0.5552	0.5576	0.5727	0.7306	0.5663	*0.5538*	0.5833	0.5698	0.5671	**0.5477**	0.5501
100	*0.5424*	0.5451	0.5599	0.7004	0.5550	0.5443	0.5731	0.5607	0.5556	**0.5368**	0.5392
120	*0.5354*	0.5371	0.5488	0.6652	0.5490	0.5385	0.5659	0.5542	0.5492	**0.5301**	0.5318
150	*0.5242*	0.5248	0.5396	0.6394	0.5381	0.5297	0.5595	0.5474	0.5386	**0.5202**	0.5217
180	*0.5193*	0.5201	0.5343	0.6180	0.5334	0.5266	0.5571	0.5454	0.5338	**0.5164**	0.5177
200	*0.5154*	0.5156	0.5316	0.6116	0.5294	0.5229	0.5532	0.5422	0.5298	**0.5125**	0.5138

Table 4: Values of $\mathrm{MSE}_{\bar{y}} \times 100$ for various methods in Empirical Application 1

n	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1	SMA2
30	2.4335	2.4058	2.3935	2.6566	2.3780	*2.3022*	2.4620	2.3498	2.3741	**2.2935**	2.2853
50	1.3180	1.3377	1.3358	1.4918	1.3169	*1.2838*	1.3738	1.3303	1.3176	**1.2745**	1.2807
80	0.7708	0.7684	0.7763	0.8482	0.7929	*0.7666*	0.8352	0.8037	0.7935	**0.7529**	0.7581
100	0.5486	0.5536	0.5749	0.6158	0.5547	*0.5461*	0.5942	0.5750	0.5567	**0.5365**	0.5344
120	*0.5177*	0.5181	0.5284	0.5652	0.5387	0.5263	0.5782	0.5580	0.5386	**0.5073**	0.5071
150	*0.3534*	0.3541	0.3684	0.3743	0.3657	0.3618	0.3982	0.3845	0.3676	**0.3472**	0.3448
180	*0.3166*	0.3183	0.3234	0.3322	0.3300	0.3265	0.3603	0.3478	0.3312	**0.3135**	0.3114
200	*0.2760*	0.2770	0.2826	0.2900	0.2899	0.2838	0.3078	0.2994	0.2900	**0.2703**	0.2682

5.3 Empirical application 2

In this subsection, we compare the various methods based on a chemical sensors dataset (Fonollosa et al. 2015, Wang et al. 2019). This dataset contains the readings of 16 chemical sensors exposed to the mixture of Ethylene and Methane at varying concentrations in air, and is to develop and test strategies to solve a wide variety of tasks, e.g., to develop algorithms for continuously monitoring or improving response time of sensory systems. More information about the dataset can be found in Fonollosa et al. (2015).

Following Wang et al. (2019), we use the readings from the last sensor as the response and those from the other sensors as covariates. Readings from the second sensor are deleted in the analysis because about 20% of the values are negative. Then, the remaining 14 covariates are arranged in descending order of their correlations with the response variable. Since trace concentrations often have a lognormal distribution (Goodson 2011), we take a log-transformation of the sensors readings. To reduce computational complexity, we compare the effectiveness of various methods using $N = 10000$ data points recorded between 200.5 and 300.5 seconds.

Table 5: Values of MSE_y for various methods in Empirical Application 2

n	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1
50	9.811E-05	7.491E-03	2.805E-05	1.817E-03	2.536E-05	2.445E-05	2.452E-05	*2.392E-05*	2.515E-05	**2.369E-05**
100	7.857E-05	4.121E-03	2.260E-05	1.743E-03	2.117E-05	*2.079E-05*	2.155E-05	2.115E-05	2.121E-05	**2.049E-05**
150	7.144E-05	2.956E-03	2.096E-05	1.719E-03	1.986E-05	*1.969E-05*	2.066E-05	2.031E-05	1.989E-05	**1.950E-05**
200	6.403E-05	2.519E-03	2.012E-05	1.709E-03	1.921E-05	*1.914E-05*	2.013E-05	1.981E-05	1.923E-05	**1.902E-05**
250	5.946E-05	2.427E-03	1.963E-05	1.699E-03	1.886E-05	*1.882E-05*	1.970E-05	1.945E-05	1.886E-05	**1.875E-05**
300	5.558E-05	2.167E-03	1.931E-05	1.693E-03	1.864E-05	*1.862E-05*	1.936E-05	1.914E-05	1.864E-05	**1.857E-05**

We set the first-order inclusion probabilities to be proportional to the first covariate to implement πPS sampling. As in Subsection 5.1, we compare methods (a)–(j). Let n be varied among $\{50, 100, 150, 200, 250, 300\}$. We consider the nested model framework, with a total of 14 candidate models. Repeatedly calculating D =1000 times, Tables 5 and 6 report

Table 6: Values of $\text{MSE}_{\bar{y}}$ for various methods in Empirical Application 2

n	SCAD	ALASSO	AGG1	AGG2	AIC	SAIC	BIC	SBIC	Cp	SMA1
50	2.092E-06	1.413E-04	5.965E-07	3.252E-05	5.197E-07	4.990E-07	5.047E-07	*4.938E-07*	5.173E-07	**4.900E-07**
100	8.390E-07	3.710E-05	2.297E-07	1.518E-05	2.155E-07	*2.122E-07*	2.223E-07	2.184E-07	2.171E-07	**2.107E-07**
150	5.434E-07	1.877E-05	1.410E-07	1.084E-05	1.325E-07	*1.317E-07*	1.416E-07	1.386E-07	1.336E-07	**1.312E-07**
200	3.514E-07	1.298E-05	9.685E-08	8.494E-06	9.315E-08	*9.289E-08*	9.539E-08	9.464E-08	9.325E-08	**9.198E-08**
250	2.366E-07	1.055E-05	7.506E-08	7.406E-06	7.258E-08	*7.202E-08*	7.467E-08	7.334E-08	7.267E-08	**7.148E-08**
300	1.670E-07	6.212E-06	6.547E-08	4.952E-06	6.262E-08	*6.236E-08*	6.442E-08	6.359E-08	6.266E-08	**6.212E-08**

the MSE_y and $\text{MSE}_{\bar{y}}$ values for the different methods, respectively. The results show that the proposed method performs best, and the model averaging methods (SAIC, SBIC, and SMA1) generally outperform their model selection counterparts (AIC, BIC, and C_p).

6 Conclusion

In this paper, we have developed a frequentist model averaging framework for analyzing complex survey data. We have proposed a Mallows-type model averaging method to estimate the mean function in the superpopulation model. Our weight selection criterion is approximately unbiased and consistent. We have established the asymptotic optimality of the resulting model averaging estimator, and derived the convergence rate of the estimated weight vector toward the theoretically optimal weight vector. When there are one or more correct models in candidate set, the total weight assigned to these correct models will converge to one. Based on the proposed model averaging method, we have constructed a model-assisted averaging estimator of the finite population total and established its theoretical properties. Numerical studies have demonstrated the effectiveness of the proposed methods.

The theoretical framework developed in this paper for complex survey data analysis may be extended to cover more general superpopulation models and prediction methods, including semiparametric and nonparametric models (Zhang & Wang 2019, Zhu et al. 2019, Lv & Ma 2026) and machine learning methods (Chen et al. 2024, Ahrens et al.

2025). Although the proposed method allows the covariate dimension to diverge, its direct application to high-dimensional settings may be limited without additional dimensionality-reduction techniques. One possible solution is to incorporate a preliminary feature screening or grouping step, as commonly adopted in high-dimensional model averaging (Ando & Li 2014, Xie et al. 2021), before constructing the candidate models. Moreover, the proposed model averaging approach may be extended to other areas of survey data analyses, such as small area estimation (Rao & Molina 2015), and to related statistical problems, including subsampling for big data (Wang et al. 2018) and missing data analysis (Shao & Wang 2016).

7 Acknowledgments

The authors thank the editor, the associate editor and the two referees for their constructive comments and suggestions that have substantially improved the original manuscript.

8 Supplementary Materials

Proofs of all theoretical results and additional numerical analyses are provided in the Supplementary Material.

9 Disclosure Statement

No potential conflict of interest was reported by the authors.

10 Data Availability Statement

The data that support the findings of this study are openly available at https://weatherspark.com and https://archive.ics.uci.edu/dataset/322.

References

Ahrens, A., Hansen, C. B., Schaffer, M. E. & Wiemann, T. (2025), 'Model averaging and double machine learning', *Journal of Applied Econometrics* **40**, 249–269.

Aitkin, M., Liu, C. C. & Chadwick, T. (2009), 'Bayesian model comparison and model averaging for small-area estimation', *The Annals of Applied Statistics* **3**, 199–221.

Ando, T. & Li, K. C. (2014), 'A model-averaging approach for high-dimensional regression', *Journal of the American Statistical Association* **109**, 254–265.

Berger, Y. G. (1996), 'Rate of convergence to normal distribution for the Horvitz-Thompson estimator', *Journal of Statistical Planning and Inference* **67**, 209–226.

Boistard, H., Lopuhaä, H. P. & Ruiz-Gazen, A. (2012), 'Approximation of rejective sampling inclusion probabilities and application to high order correlations', *Electronic Journal of Statistics* **6**, 1967–1983.

Breidt, F. J. & Opsomer, J. D. (2000), 'Local polynomial regression estimators in survey sampling', *The Annals of Statistics* **28**, 1026–1053.

Breidt, F. J., Opsomer, J. D., Johnson, A. A. & Ranalli, M. G. (2007), 'Semiparametric model-assisted estimation for natural resource surveys', *Survey Methodology* **33**, 35–44.

Buckland, S. T., Burnham, K. P. & Augustin, N. H. (1997), 'Model selection: An integral part of inference', *Biometrics* **53**, 603–618.

Bunea, F., Tsybakov, A. B. & Wegkamp, M. H. (2007), 'Aggregation for gaussian regression', *The Annals of Statistics* **35**, 1674–1697.

Cardot, H., Chaouch, M., Goga, C. & Labruère, C. (2010), 'Properties of design-based functional principal components analysis', *Journal of Statistical Planning and Inference* **140**, 75–91.

Cassel, C.-E., Särndal, C.-E. & Wretman, J. H. (1976), 'Some results on generalized difference estimation and generalized regression estimation for finite populations', *Biometrika* **63**, 615–620.

Cassel, C.-M., Särndal, C.-E. & Wretman, J. H. (1977), *Foundations of Inference in Survey Sampling*, John Wiley & Sons, New York.

Chen, X., Yu, D. & Zhang, X. (2024), 'Optimal weighted random forests', *Journal of Machine Learning Research* **25**, 1–81.

Claeskens, G. & Hjort, N. L. (2008), *Model Selection and Model Averaging*, Cambridge University Press, Cambridge.

Clyde, M. A. & Ghosh, J. (2012), 'Finite population estimators in stochastic search variable selection', *Biometrika* **99**, 981–988.

Cochran, W. G. (1977), *Sampling Techniques*, 3rd edn, John Wiley & Sons, New York.

Delevoye, A. & Sävje, F. (2020), 'Consistency of the Horvitz-Thompson estimator under general sampling and experimental designs', *Journal of Statistical Planning and Inference* **207**, 190–197.

Elliott, M. R. (2008), 'Model averaging methods for weight trimming', *Journal of Official Statistics* **24**, 517–540.

Elliott, M. R. (2009), 'Model averaging methods for weight trimming in generalized linear regression models', *Journal of Official Statistics* **25**, 1–20.

Fan, J. & Li, R. (2001), 'Variable selection via nonconcave penalized likelihood and its oracle properties', *Journal of the American Statistical Association* **96**, 1348–1360.

Fang, F., Lan, W., Tong, J. & Shao, J. (2019), 'Model averaging for prediction with fragmentary data', *Journal of Business and Economic Statistics* **37**, 517–527.

Fonollosa, J., Sheik, S., Huerta, R. & Marco, S. (2015), 'Reservoir com-puting compensates slow response of chemosensor arrays exposedto fast varying gas concentrations in continuous monitoring', *Sensors and Actuators B: Chemical* **215**, 618–629.

Gao, Y., Zhang, X., Wang, S. & Zou, G. (2016), 'Model averaging based on leave-subject-out cross-validation', *Journal of Econometrics* **192**, 139–151.

Goodson, D. Z. (2011), *Mathematical Methods for Physical and Analytical Chemistry*, Wiley, New York.

Gregoire, T. G. (1998), 'Design-based and model-based inference in survey sampling: Appreciating the difference', *Canadian Journal of Forest Research* **28**, 1429–1447.

Hájek, J. (1959), 'Optimal strategy and other problems in probability sampling', *Časopis pro Pěstování Matematiky* **84**, 387–423.

Hájek, J. (1960), 'Limiting distributions in simple random sampling from a finite population', *Magyar Tudományos Akadémia Matematikai Kutató Intézetének Közleményei* **5**, 361–374.

Hájek, J. (1964), 'Asymptotic theory of rejective sampling with varying probabilities from a finite population', *Annals of Mathematical Statistics* **35**, 1491–1523.

Hansen, B. E. (2007), 'Least squares model averaging', *Econometrica* **75**, 1175–1189.

Hansen, B. E. & Racine, J. (2012), 'Jackknife model averaging', *Journal of Econometrics* **167**, 38–46.

Hansen, M. H., Madow, W. G. & Tepping, B. J. (1983), 'An evaluation of model-dependent and probability-sampling inferences in sample surveys', *Journal of the American Statistical Association* **78**, 776–793.

Hjort, N. & Claeskens, G. (2003), 'Frequentist model average estimators', *Journal of the American Statistical Association* **98**, 879–899.

Horvitz, D. G. & Thompson, D. J. (1952), 'A generalization of sampling without replacement from a finite universe', *Journal of the American Statistical Association* **47**, 663–685.

Isaki, C. T. & Fuller, W. A. (1982), 'Survey design under the regression superpopulation model', *Journal of the American Statistical Association* **77**, 89–96.

Leung, G. & Barron, A. R. (2006), 'Information theory and mixing least-squares regressions', *IEEE Transactions on Information Theory* **52**, 3396–3410.

Li, C., Li, Q., Racine, J. S. & Zhang, D. (2018), 'Optimal model averaging of varying coefficient models', *Statistica Sinica* **28**, 2795–2809.

Li, X. & Opsomer, J. D. (2006), Model averaging in survey estimation, *in* 'Proceedings of the Survey Research Methods Section', American Statistical Association, Alexandria, VA.

Liang, F. & Kuk, A. Y. C. (2004), 'A finite population estimation study with Bayesian neural networks', *Survey Methodology* **30**, 219–234.

Liang, H., Zou, G., Wan, A. T. K. & Zhang, X. (2011), 'Optimal weight choice for frequentist model average estimators', *Journal of the American Statistical Association* **106**, 1053–1066.

Liang, X., Li, S., Zhang, S., Huang, H. & Chen, S. (2016), 'PM2.5 data reliability, consistency, and air quality assessment in five Chinese cities', *Journal of Geophysical Research: Atmospheres* **121**, 10220–10236.

Liao, J., Zong, X., Zhang, X. & Zou, G. (2019), 'Model averaging based on leave-subject-out cross-validation

for vector autoregressions', *Journal of Econometrics* **209**, 35–60.

Liao, J. & Zou, G. (2020), 'Corrected Mallows criterion for model averaging', *Computational Statistics & Data Analysis* **144**, 106902.

Liu, Z., Zheng, J., Tu, C. & Pan, Y. (2023), 'Estimation for volunteer web survey samples using a model-averaging approach', *Journal of Applied Statistics* **50**, 3251–3271.

Lu, X. & Su, L. (2015), 'Jackknife model averaging for quantile regressions', *Journal of Econometrics* **188**, 40–58.

Lv, J. & Ma, S. (2026), 'Optimal conditional quantile prediction via model averaging of partially linear additive models', *Statistica Sinica* **36**, 147–169.

Nemirovski, A. (2000), Topics in non-parametric statistics, *in* 'Lectures on Probability Theory and Statistics', Lecture Notes in Mathematics, Springer, Berlin.

Newey, W. (1997), 'Convergence rates and asymptotic normality for series estimators', *Journal of Econometrics* **79**, 147–168.

Raghunath, A. (1990), 'On comutativity of design and model expectations in randomized response surveys', *Communications in Statistics - Theory and Methods* **19**, 3751–3757.

Rao, J. N. K. & Molina, I. (2015), *Small Area Estimation*, 2 edn, John Wiley & Sons, Hoboken.

Robinson, P. M. & Särndal, C.-E. (1983), 'Asymptotic properties of the generalized regression estimator in probability sampling', *Sankhyā: Indian Journal of Statistics, Series B* **45**, 240–248.

Särndal, C.-E. (1980), 'On π-inverse weighting versus best linear unbiased weighting in probability sampling', *Biometrika* **67**, 639–650.

Särndal, C.-E., Swensson, B. & Wretman, J. (1992), *Model Assisted Survey Sampling*, Springer-Verlag, New York.

Särndal, C.-E., Thomsen, I., Hoem, J. M., Lindley, D. V., Barndorff-Nielsen, O. & Dalenius, T. (1978), 'Design-based and model-based inference in survey sampling [with discussion and reply]', *Scandinavian Journal of Statistics* **5**, 27–52.

Shao, J. & Wang, L. (2016), 'Semiparametric inverse propensity weighting for nonignorable missing data', *Biometrika* **103**, 175–187.

Ta, T., Shao, J., Li, Q. & Wang, L. (2020), 'Generalized regression estimators with high-dimensional covariates', *Statistica Sinica* **30**, 1135–1154.

Tsybakov, A. B. (2003), Optimal rates of aggregation, *in* 'Learning Theory and Kernel Machines', Lecture Notes in Artificial Intelligence, Springer, Heidelberg.

Valliant, R., Dorfman, A. H. & Royall, R. M. (2000), *Finite Population Sampling and Inference: A Prediction Approach*, John Wiley & Sons, New York.

Wan, A. T. K., Zhang, X. & Zou, G. (2010), 'Least squares model averaging by Mallows criterion', *Journal of Econometrics* **156**, 277–283.

Wang, H., Yang, M. & Stufken, J. (2019), 'Information-based optimal subdata selection for big data linear regression', *Journal of the American Statistical Association* **114**, 393–405.

Wang, H., Zhu, R. & Ma, P. (2018), 'Optimal subsampling for large sample logistic regression', *Journal of the American Statistical Association* **113**, 829–844.

Wang, H., Zou, G. & Wan, A. T. K. (2012), 'Model averaging for varying-coefficient partially linear measurement error models', *Electronic Journal of Statistics* **6**, 1017–1039.

Wang, L. (2009), 'Single-index model-assisted estimation in survey sampling', *Journal of Nonparametric Statistics* **21**, 487–504.

Wang, L. & Wang, S. (2011), 'Nonparametric additive model-assisted estimation for survey data', *Journal of Multivariate Analysis* **102**, 1126–1140.

Xie, J., Yan, X. & Tang, N. (2021), 'A model-averaging method for high-dimensional regression with missing responses at random', *Statistica Sinica* **31**, 1005–1026.

Yang, Y. (2000), 'Combining different procedures for adaptive regression', *Journal of Multivariate Analysis* **74**, 135–161.

Yang, Y. (2001), 'Adaptive regression by mixing', *Journal of the American Statistical Association* **96**, 574–588.

Zhang, X., Chiou, J. & Ma, Y. (2018), 'Functional prediction through averaging estimated functional linear regression models', *Biometrika* **105**, 945–962.

Zhang, X., Ma, Y. & Carroll, R. J. (2019), 'MALMEM: Model averaging in linear measurement error models', *Journal of the Royal Statistical Society (Series B)* **81**, 763–779.

Zhang, X., Wan, A. T. K. & Zou, G. (2013), 'Model averaging by jackknife criterion in models with dependent data', *Journal of Econometrics* **174**, 82–94.

Zhang, X. & Wang, W. (2019), 'Optimal model averaging estimation for partially linear models', *Statistica Sinica* **29**, 693–718.

Zhang, X., Yu, D., Zou, G. & Liang, H. (2016), 'Optimal model averaging estimation for generalized linear models and generalized linear mixed-effects models', *Journal of the American Statistical Association* **111**, 1775–1790.

Zhang, X., Zou, G. & Carroll, R. (2015), 'Model averaging based on Kullback-Leibler distance', *Statistica Sinica* **25**, 1583–1598.

Zhang, X., Zou, G. & Liang, H. (2014), 'Model averaging and weight choice in linear mixed-effects models', *Biometrika* **101**, 205–218.

Zhu, R., Wan, A. T. K., Zhang, X. & Zou, G. (2019), 'A mallows-type model averaging estimator for the varying-coefficient partially linear model', *Journal of the American Statistical Association* **114**, 882–892.

Zong, X., Tso, G. & Zou, G. (2026), 'Improved model-assisted estimation via probability thresholding', *Statistica Sinica* **36**, 53–78.

Zong, X., Zhu, R. & Zou, G. (2019), 'Improved Horvitz-Thompson estimator in survey sampling', *Survey Methodology* **45**, 165–184.

Zou, H. (2006), 'The adaptive lasso and its oracle properties', *Journal of the American Statistical Association* **101**, 1418–1429.